

Comparing Two Typing Methods

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1 Introduction

In this study, we are going to investigate the feasibility of swiping across a soft keyboard as a one-handed input method, as opposed to pressing the individual keys. Swiping can be used as an input method when a user swipes across the keyboard in one motion, only pausing on the letters they need to input, and only lifting their finger when they have spelled the entire word. Pressing individual keys can require more precise movement concerning the hand than might be possible for users, especially concerning key size. Since swiping has a built-in algorithm for predicting the word the user wants to spell and does not completely rely on the user being accurate in their movements, this study aims to investigate in what way this input method can be an accessibility feature when compared to pressing individual keys on a soft keyboard.

This study aims to answer two research questions, one for qualitative research and one for quantitative research. As qualitative research, this study aims to investigate in what way swiping across keys rather than pressing keys on a soft keyboard influences the accuracy and speed of inputting words as a one-handed input method. This study also aims to investigate in what way swiping across keys rather than pressing the keys influences the user experience regarding ease of use and the experience of being efficient and effective while typing. The measure for speed in this study is raw words per minute. This is the total number of characters inputted divided by 5 and then normalized to 60 seconds. This measure is used because on average the words in the software used had 5 letters. The accuracy is the percentage of correctly pressed keys. To measure the user feedback this study uses the System Usability Scale, with some edited questions.

We expect that due to inexperience using the swiping users have a higher speed while typing, but because they need to concentrate more on the swiping rather than the input method being instinctive as is with pressing the keys, the accuracy of the swiping is higher than pressing the keys. We also expect the user experience to be more positive for pressing the keys than for the swiping, mostly because of habit. As such, we expect users to find themselves less efficient and less effective when using swiping as an input method than pressing keys as an input method.

2 Experiment

This experiment combines questionnaires and typing tests using two different typing methods. Student participants are required to fill in questionnaires and receive two one-hand typing tests in a quiet meeting room. In total we tested with 3 participants. The dependent variables in this experiment were the typing speed in words per minute and the accuracy. The independent variable was the input method. The controlled variables were the software and hardware used for the experiment, along with the position of the participant and lighting of the room to control for readability of the words. The participants were also only allowed to use one hand to input the words. The typing tests are implemented through a web application called Monkeytype[1] as software. The test was executed on an iPhone 14 Pro Max and iOS 17.6.1 as hardware.

2.1 Monkeytype

Monkeytype is a minimalistic and customizable typing test. The user can edit their typing test in a large number of ways. For the test with the specified settings, Monkeytype then randomly assigns words for the user to input. These words are 200 most common English words. After the test is finished, Monkeytype then presents the stats back to the user, including typing speed and accuracy at the end of a test. For this experiment the settings of Monkeytype were left on default except for the time limit. This was either put at one minute or at two minutes depending on the phase of the experiment.

2.2 Typing methods

Participants of our experiment used two different typing methods to do the test.

- *Keypress method.* Users can only enter one letter in one input, and multiple keystrokes are required to enter a complete word.
- *Swipe method.* From the moment the finger touches the screen to the moment it leaves the screen is considered a valid input. The algorithm will calculate the word entered by the user based on the sliding path. One input can enter a complete word.

2.3 Procedure

After giving consent about the collecting data, the participants were required to do two rounds of testing, one with each input method, swipe and keypress. They were first given two minutes to practice with the software and to get used to the input method. These scores were not recorded. After these two minutes, the participants would

do another typing test using the same method for one minute. The results of this test were recorded. After the test, the participants were required to fill out the System Usability Scale questionnaire with two edited questions: instead of statements 4 and 5 "I think that I would need the support of a technical person to be able to use this system" and "I found the various functions in this system were well integrated," the statements "I thought I was accurate using the system" and "I thought I was typing quickly" were added.

After this first round, the participants would then redo the experiment but then with the other input method, since we have chosen for a within-subjects design. The order of these rounds was randomly decided, but it was made sure that both orders were tested as evenly as possible.

During the experiment, the participants were deliberately not instructed to concentrate on either speed or accuracy, and left it to their own discretion whether they would put accuracy above speed - by for example removing incorrect inputs - or whether they would continue to the next word immediately after an incorrect input. This was so that the participant would use the input method as naturally as possible without external interference.

2.4 Measure

We used a combination of questionnaires and typing tests to measure differences between participants and their experience with the two typing methods.

Quantitative measure. The quantitative measures can reflect directly the quality of typing in each test.

- *Speed.* This is measured through words per minute (WPM). The WPM in Monkeytype is the total number of characters in the typed words (including spaces), divided by 5 and normalized to 60 seconds.
- *Accuracy.* The accuracy in a test is the percentage of correctly pressed keys.

Qualitative measure. The qualitative measure is implemented through a usability questionnaire. Each participant needs to fill in the questionnaire after finishing each typing test. By analyzing the results, we can estimate the user experience of typing using different methods.

2.5 Limitation

Since we only have 3 experimental samples, it may bias the results because it may not capture the diversity and variability that exists in the entire population. It is also the case that one out of three participants can only use one hand, and as such the data can be skewed due to this, because this participant might have been more experienced in only using a single hand as an input method. Our population sample was also only men, resulting in results that do not properly represent the whole population. Since the participants were not instructed to concentrate on either speed or accuracy, this study does not have the data on what the participants put emphasis on. In further research it is valuable to look at the input methods when they are explicitly used to be either as accurate as possible or as fast as possible.

3 Results

3.1 Approach

After a 2 minute trial period for each method of typing, users were tested for 1 minute of typing randomly generated word prompts. We would alternate between starting with swipe and keypress typing for each user tested. The collected data is displayed in Figure 1, which shows the resulting words per minute (WPM) as well as the Accuracy (%). Finally, our qualitative testing was performed using a modified SUS questionnaire for both the swipe and keypress systems.

3.2 Qualitative Data

The following questions were used to survey each method of typing:

- (1) I think that I would like to use this system frequently.
- (2) I found the system unnecessarily complex.
- (3) I thought the system was easy to use.
- (4) I thought I was accurate while using the system.
- (5) I thought I was typing quickly.
- (6) I thought there was too much inconsistency in this system.
- (7) I would imagine that most people would learn to use this system very quickly.
- (8) I found the system very cumbersome to use.
- (9) I felt very confident using the system.
- (10) I needed to learn a lot of things before I could get going with this system.

Swipe					Keypress			
Question	U1	U2	U3	AVG	U1	U2	U3	AVG
1	4	3	3	3,33	5	4	2	4,67
2	2	2	2	2,00	1	3	1	1,67
3	4	4	4	4,00	5	3	4	4,33
4	5	4	3	4,00	5	5	5	5,00
5	4	4	3	3,67	5	4	4	4,67
6	2	2	3	2,33	1	2	3	1,33
7	4	3	4	3,67	4	5	3	4,33
8	1	3	3	2,33	1	4	3	2,00
9	4	4	4	4,00	5	5	4	5,00
10	3	2	2	2,33	1	2	2	1,33

Table 1: System Usability Scale Results

Each user's responses are recorded in the tables above, with the average rating displayed in the final column.

3.3 Quantitative Data

Plotting the experiment in Figure 1 shows two sets of data, with the swipe system data in blue and the keypress data in red. The data is ordered so that the first user datapoint of the blue set corresponds with the first of the red, and so on.

3.4 Analysis

Although our sample size is less than the ideal experiment, there is still a noticeable difference present between the swipe and the keypress data set.

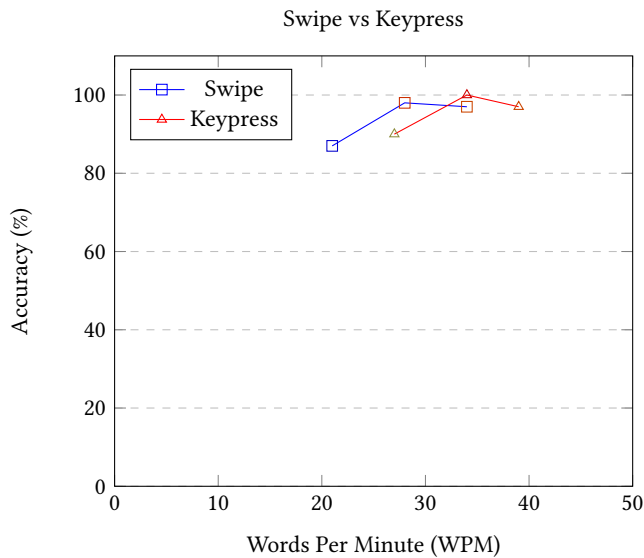


Figure 1: System Comparison

As seen in Figure 1, user testing consistently found that swiping was less efficient in both speed and accuracy. This could be attributed to the fact that swiping was often unused prior to testing by our subjects, and therefore the adjustment from a muscle memory based approach such as keypresses could play a large role. Accuracy was practically the same between each method, where each subject was able to achieve a similar word accuracy entering different text regardless of the method of typing used. Typing speed, however, varied more between the methods, with keypresses allowing the users to enter more within the same timeframe.

After completion of the experiment, each user was asked to fill out a Likert scale to review their experience using the swipe and keypress systems. One factor we wanted to focus on was how efficient or accurate the users felt while using the system. Even if the user was more efficient using the swipe according to the data, if the user reported they experienced the opposite then the system is still struggling to provide an intuitive user experience. Based on the survey, users did indeed feel as though they were less accurate - even though the data showed they were nearly the same. Efficiency wise, the results from the survey and the data align in that both showed less efficient usage.

4 Discussion

In this study we investigated both the quantitative and qualitative aspects of using the swipe method versus the traditional keypress method when typing on a soft keyboard. Although only three participants were used for our experiment and the results are thus not statistically significant, we can still try to observe potential trends and speculate about what they could imply. As noted earlier, the participants performed nearly identically in terms of accuracy for both input methods. However, the typing speed, measured in words per minute, was faster for the keypress method than it was for the swipe method. This suggests that users are able to type just as accurate with the swipe method as with the keypress method,

but that it comes at the cost of having to slow down their typing speed. This could be because the users had to be more thoughtful about their movements, thus slowing them down. Another explanation could be that the participants were simply not as experienced with the swipe method compared to the keypress method and were slower as a result. For the qualitative analysis, we used a modified version of the SUS Questionnaire. This questionnaire uses a Likert scale to assess the usability of a given system. Again, since we only used three participants, the results of this questionnaire can not tell us anything definitively about the usability of either system, but some trends can be observed. Noteworthy is that the participants noted that they would use the keypress method more frequently than the swipe method, and that the swipe method felt less accurate and slower. Interestingly, them being less accurate when using the swipe method is not supported by the quantitative data. This shows a clear difference in subjective experience versus objective performance.

Additionally, the participants found the swipe method to be more inconsistent and they felt less confident using it. Finally, they found the learning curve to be steeper. All of these findings could simply be the result of the participants having little to no experience with the swipe input method. Perhaps, if the users were given more time than just two minutes, the results would have differed greatly. Of course, it could also be the case that the swipe method is indeed less reliable, even if the users are given ample time to adjust to it.

Based on the results of our experiment the swipe input method does not seem like a promising accessibility feature, as users are able to type faster using the keypress method and also feel faster and more accurate.

It is important to consider, however, that most people have been using the traditional keypress input method for a long time. We could imagine that, in order for their confidence using the swipe method to be just as high as with the keypress method, a similarly long time might be needed. Determining the length of this adjustment period could be an interesting topic for further research, as this has great implications for the usability of the swipe input method.

Furthermore, it is important to consider the typing test used in this study. By not including punctuation and by only having relatively simple and common words appear in the text, the results were definitively influenced. We expect this to have favored the swipe input method slightly, as having to type punctuation or maybe even just longer words may take the user out of the "swiping flow", but investigating this further could be a promising area for further research. The same goes for the effect of the word complexity on the performance for both input methods.

Finally, a perhaps logical follow-up study would be to replicate this study in similar fashion, but with a higher number of participants, allowing for a more refined statistical analysis.

5 Conclusion

In this study we investigated in what way the use of either the traditional keypress or the swipe input method impacted the performance of the user in terms of typing speed and accuracy, as well as their user experience in terms of usability and perceived efficiency.

We did this by running an experiment where the participants performed a typing test using both input methods, after which filling in a questionnaire. Although no statistically significant conclusions can be drawn, our quantitative results suggest that users were able to type faster using the keypress method, whilst having similar accuracy for both input methods. However, the qualitative results not only imply a loss in perceived speed when using the swipe method compared to the keypress method, but also a loss in perceived accuracy, thus leading to a less efficient user experience. This leads us to believe the swipe method does not show promise to be a efficient accessibility feature, although it is important to

consider the difference in experience with the two input methods the participants likely had.

6 Task distribution

For this assignment we each spent around 12 hours for writing the assignment, holding a meeting twice and doing the experiment. As for writing this assignment, the work was split equally.

References

- [1] 2024. Monkeytype | A minimalistic, customizable typing test. <https://monkeytype.com> [Online; accessed 9. Oct. 2024].